JKO-Based Framework for a Nonlinear Fokker-Planck Type Model of Grain Growth

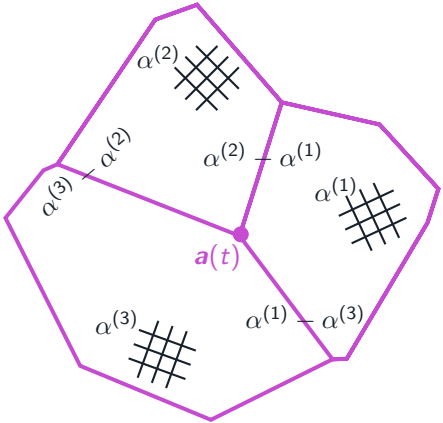
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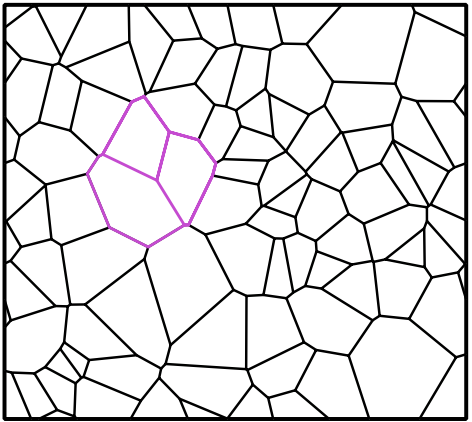
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What is grain growth?

- Grain growth is a highly complex multiscale-multiphysics process appearing in materials science which describes the evolution of the microstructure of polycrystalline materials
- These materials consist of many small monocrystalline grains which are separated by grain boundaries.
- The changes in grain size affect the material's electrical, thermal, etc. properties, which are important in the engineering of new materials.

Grain boundary networks and statistical observables





- $\boldsymbol{a}(t)$ is the triple junction point,
- $\alpha^{(j)} = \alpha^{(j)}(t)$'s are lattice orientations,
- $\alpha^{(i)} - \alpha^{(j)}$'s are lattice misorientations.

Vertex model

$N \to \infty$ limit of the vertex model resulting in the following free energy with inhomogeneous absolute temperature $D(x)$:

$$F[f] = \int_\Omega \left(D(x) f(x,t)(\log f(x,t) - 1) + \phi(x) f(x,t) \right) dx$$

together with the dissipation relation

$$\frac{d}{dt} F[f] = -\int_\Omega \frac{f}{\pi(x)} \left| \nabla (D(x) \log f + \phi(x)) \right|^2 dx.$$

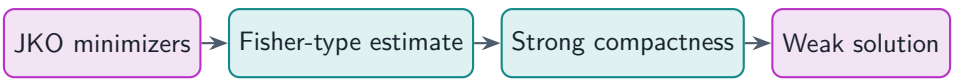
Nonlinear Fokker-Planck type model

$$\begin{cases} \dfrac{\partial f}{\partial t} = \nabla \cdot \left(\dfrac{f}{\pi(x)} \nabla \left(D(x) \log f + \phi(x) \right) \right), & x \in \Omega, \ t \in (0, \infty), \\ \dfrac{f}{\pi(x)} \nabla \left(D(x) \log f + \phi(x) \right) \cdot \nu = 0, & x \in \partial\Omega, \\ f(x, 0) = f_0(x), & x \in \Omega. \end{cases} \tag{nFP}$$

- $f(x, t)$ will be a probability density function, think $x = (a, \Delta\alpha) \in \Omega \subset \mathbb{R}^d$,
- $\pi(x) > 0$ is mobility function,
- $D(x) > 0$ is absolute temperature function,
- $\phi(x)$ is energy density of a grain boundary.

Main results

1. **Global weak solutions** exists under natural (no-flux) boundary conditions for a nonnegative finite free energy initial data.
2. **Discrete maximum principle** for the JKO minimizers.
3. **Exponential convergence to the equilibrium** for smooth solutions of the model.



Linear Fokker–Planck as a Wasserstein gradient flow

Let

$$E[\rho] := \int_\Omega \rho \log \rho \, dx + \int_\Omega V(x)\rho(x) \, dx, \quad \text{and} \quad W_2^2(\mu_1, \mu_2) := \inf_{\gamma \in \Gamma(\mu_1, \mu_2)} \left\{ \int_{\Omega \times \Omega} |x - y|^2 \, d\gamma \right\}.$$

For a given time step size $\tau > 0$, consider the minimizing movements scheme [JKO]

$$\rho_{k+1}^\tau \in \operatorname*{argmin}_{\rho \in \mathcal{P}(\Omega)} \left\{ \frac{W_2^2(\rho, \rho_k^\tau)}{2\tau} + E[\rho] \right\}. \tag{JKO}$$

Define $\rho^\tau(t) := \rho_k^\tau, \ t \in ((k-1)\tau, k\tau]$. Jordan, Kinderlehrer, and Otto proved that, as $\tau \to 0$, ρ^τ weakly converges to the weak solution ρ of the linear Fokker-Planck $\frac{\partial \rho}{\partial t} = \Delta \rho + \nabla \cdot (\rho \nabla V)$.

[JKO] The Variational Formulation of the Fokker–Planck Equation, Jordan, Kinderlehrer, and Otto, 1998, SIMA

Assumptions

Domain and initial data

- $\Omega \subset \mathbb{R}^d$ is bounded, connected, and C^1 domain.
- $f_0 \geq 0$, and $\int_\Omega f_0 dx = 1$.
- $F[f_0] < \infty$.

Coefficients

- $D \in C^2(\Omega)$ and $D \geq C_D > 0$.
- $\pi \in C^1(\Omega)$ and $0 < C_\pi^{\text{low}} \leq \pi \leq C_\pi^{\text{up}}$.
- $\phi \in C^1(\Omega)$.

Important relaxation Our weak existence theorem does not require $f_0 \geq c > 0$, as opposed to [2].

[2] Batuhan Bayir, Yekaterina Epshteyn and William M Feldman, Global well-posedness of a nonlinear Fokker-Planck type model of grain growth, *Discrete and Continuous Dynamical Systems (DCDS)*, 2026.

Mobility-weighted Wasserstein distance

$$c_\pi(x,y) := \pi(y)|x-y|^2,$$
$$W_\pi^2(\mu_1,\mu_2) := \inf_{\gamma\in\Gamma(\mu_1,\mu_2)} \left\{ \int_{\Omega\times\Omega} \pi(y)|x-y|^2 \, d\gamma \right\}.$$

Equivalence with W_2 $\quad C_\pi^{\mathrm{low}} W_2^2 \le W_\pi^2 \le C_\pi^{\mathrm{up}} W_2^2.$

W_π is generally a quasi distance because the cost uses $\pi(y)$, but it induces the same compactness properties as W_2 on bounded Ω.

The JKO scheme

Let $\tau > 0$ be a given time step size, and $f_0^\tau := f_0$. Then, for a given f_k^τ we recursively define f_{k+1}^τ as the minimizer of

$$f_{k+1}^\tau \in \operatorname*{argmin}_{f \in \mathcal{P}(\Omega)} \left\{ \frac{W_\pi^2(f, f_k^\tau)}{2\tau} + F[f] \right\}. \qquad (\mathrm{JKO}_{\mathrm{nFP}})$$

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Lemma

At each step $k \geq 1$, there exists a unique minimizer of $(\mathrm{JKO}_{\mathrm{nFP}})$ over $\mathcal{P}(\Omega)$.

Euler-Lagrange equation

For every smooth tangential vector field ξ,

$$\frac{1}{\tau}\int_{\Omega\times\Omega}\pi(y)\xi(x)\cdot(x-y)\,d\gamma$$
$$=\int_\Omega Df_k^\tau\,\nabla\cdot\xi dx-\int_\Omega f_k^\tau(\log f_k^\tau-1)\nabla D\cdot\xi dx-\int_\Omega f_k^\tau\nabla\phi\cdot\xi dx.$$

Moreover, for a Kantorovich potential φ_k,

$$D(x)\log f_k^\tau+\phi(x)+\frac{\varphi_k(x)}{\tau}=C_k\qquad\text{a.e.}$$

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Goal Obtain strong L^1 convergence of both f^τ and $f^\tau\log f^\tau$.

Idea of the proof

Idea: Control the dissipation by the Fisher information $\mathcal{I} := \int_\Omega \frac{|\nabla f|^2}{f} dx = 4 \left\| \nabla \sqrt{f} \right\|_{L^2(\Omega)}$.

For an optimal map T_k from f_{k-1}^τ to f_k^τ, we have

$$\begin{aligned} W_\pi^2(f_k^\tau, f_{k-1}^\tau) &= \frac{\tau^2}{4} \int_\Omega \frac{f_k^\tau}{\pi(\mathrm{T}_k(x))} \left| \nabla (D(x) \log f_k^\tau + \phi(x)) \right|^2 dx \\ &\geq \frac{\tau^2}{4 C_\pi^{\mathrm{up}}} \mathrm{Diss}(f_k^\tau), \end{aligned}$$

where

$$\begin{aligned} \mathrm{Diss}(f) = &\int_\Omega (D(x))^2 \frac{|\nabla f|^2}{f} dx + \int_\Omega 2D(x) \nabla D(x) \log f \cdot \nabla f dx \\ &+ \int_\Omega 2(\nabla D(x) \cdot \nabla \phi(x)) f \log f dx + \int_\Omega 2D(x) \nabla \phi(x) \cdot \nabla f dx \\ &+ \int_\Omega |\nabla \phi(x)|^2 f dx + \int_\Omega |\nabla D(x)|^2 f (\log f)^2 dx \\ =: &\, I_1 + I_2 + I_3 + I_4 + I_5 + I_6. \end{aligned}$$

Idea of the proof (continued)

Since $I_5, I_6 \geq 0$, and $D(x) \geq C_D > 0$,

$$\mathrm{Diss}(f) \geq C_D^2 \mathcal{I} + I_2 + I_3 + I_4, \qquad \mathcal{I} = \mathcal{I}[f] := \int_\Omega \frac{|\nabla f|^2}{f} dx = 4 \left\| \nabla \sqrt{f} \right\|_{L^2(\Omega)}.$$

Denote $\mathrm{Ent}(f) := \int_\Omega f \log f dx$. It remains to prove

$$|I_2| + |I_3| + |I_4| \leq \frac{3}{4} C_D^2 \mathcal{I} + C\big(\mathrm{Ent}(f) + 1\big).$$

Then, this will imply Fisher-type spacetime regularity i.e., $\sqrt{f^\tau} \in L_t^2 H_x^1$.

Idea of the proof (continued)

To estimate the integral I_2, we integrate by parts and get

$$\begin{aligned} I_2 &= \int_\Omega \nabla(D(x)^2)\log f \cdot \nabla f dx \\ &= -\int_\Omega f\log f \Delta(D(x)^2)dx - \int_\Omega \nabla f \cdot \nabla(D(x)^2)dx \\ &\quad + \int_{\partial\Omega} f\log f \nabla(D(x)^2)\cdot \nu dS \\ &=: I_{2,1} + I_{2,2} + I_{2,3}. \end{aligned}$$

By Young's inequality,

$$|I_{2,1}| + |I_{2,2}| + |I_3| + |I_4| \le \frac{C_D^2}{2}\mathcal{I} + C(\mathrm{Ent}(f) + 1).$$

Idea of the proof (continued)

The boundary integral $I_{2,3}$ requires a more careful estimation.

First, we derive a trace inequality with ε. For any $u \in H^1(\Omega)$ and $\varepsilon > 0$,

$$\|u\|^q_{L^q(\partial\Omega)} \leq \varepsilon \|\nabla u\|_2^2 \|u\|_2^{q-2} + C_\varepsilon \|u\|_2^q, \quad 2 \leq q < 2 + 2/d.$$

Second, we let $u \coloneqq \sqrt{f}$, then for any $2 < q$, we have

$$|f \log f| = 2u^2 \, |\log u| \leq C(q) u^q + 1.$$

Combining these two gives,

$$|I_{2,3}| = \left| \int_{\partial\Omega} f \log f \nabla (D(x)^2) \cdot \nu dS \right| \leq \frac{C_D^2}{4} \mathcal{I} + C.$$

Fisher-type spacetime regularity

For any $T < \infty$, the piecewise constant interpolation f^τ satisfies the following estimates

$$\left\| \sqrt{f^\tau} \right\|^2_{L^2(0,T;H^1(\Omega))}, \quad \|f^\tau\|_{L^1(0,T;W^{1,1}(\Omega))} \leq C(1+T),$$

and

$$\|f^\tau\|^{1+\frac{\alpha}{1+\alpha}}_{L^{1+\frac{\alpha}{1+\alpha}}(\Omega\times[0,T])} \leq C(1+T).$$

Here, if $d > 2$, $\alpha := \frac{2}{d-2}$. If $d \leq 2$, $\alpha \in [1, \infty)$.

- The first estimate helps to apply Aubin-Lions-Simons type lemma from Dreher and Jüngel [DJ].
- The second estimate helps to get uniform integrability of $\{f^\tau \log f^\tau\}_{\tau>0}$ via de la Vallée Poussin's criterion.

[DJ] Dreher, M., and Jüngel, A. (2012). Compact families of piecewise constant functions in $L^p(0, T; B)$. Nonlinear Analysis: Theory, Methods & Applications.

Strong compactness

For any $T < \infty$, and sequence of time steps $\{\tau_k\}_{k \geq 1}$ such that $\tau_k \to 0$ as $k \to \infty$, up to a (not relabeled) subsequence, we have

$$f^{\tau_k} \to f \text{ strongly in } L^1(\Omega \times [0, T]),$$

and

$$f^{\tau_k} \log f^{\tau_k} \to f \log f \text{ strongly in } L^1(\Omega \times [0, T]).$$

Weak solution

Let f_0 be an initial data such that $F[f_0] < \infty$, and $\|f_0\|_{L^1(\Omega)} = 1$. The function $f \in L^1_{\mathrm{loc}}(\Omega \times [0,\infty))$ is a weak solution of (nFP) if

$$
\begin{aligned}
0 = -\psi(0) \int_\Omega \zeta(x) f_0(x) dx - \int_0^\infty \int_\Omega \partial_t \psi(t) \zeta(x) f(x,t) dx dt \\
- \int_0^\infty \int_\Omega \psi(t) \frac{1}{\pi(x)} \nabla \zeta(x) \cdot \nabla D(x) f(x,t) (1 - \log f(x,t)) dx dt \\
- \int_0^\infty \int_\Omega \psi(t) \nabla \cdot \left(\frac{\nabla \zeta(x)}{\pi(x)} \right) f(x,t) dx dt + \int_0^\infty \int_\Omega \psi(t) \frac{\nabla \zeta(x)}{\pi(x)} \cdot \nabla \phi(x) f(x,t) dx dt
\end{aligned}
$$

holds for all $\psi \in C_c^\infty([0,\infty))$ and $\zeta \in H^1(\Omega)$.

Main result

Theorem

There exists a weak solution $f \in L^1_{\mathrm{loc}}(\Omega \times [0,\infty))$ *of* (nFP).

Discrete maximum principle

Proposition

Assume that

$$f_0 \in \mathcal{P}^{\mathrm{ac}}(\Omega), \ \textit{and} \ \ D(x)\log f_0(x) + \phi(x) \in L^\infty(\Omega).$$

Then, any minimizer f_{k+1}^τ *of* $(\mathrm{JKO}_{\mathrm{nFP}})$ *satisfies the following discrete maximum principle*

$$\left\| D\log f_{k+1}^\tau + \phi \right\|_{L^\infty(\Omega)} \leq \left\| D\log f_0 + \phi \right\|_{L^\infty(\Omega)}.$$

The proof inspired by the work of Tudorascu [T].

[T] On the Jordan–Kinderlehrer–Otto variational scheme and constrained optimization in the Wasserstein metric, Tudorascu, 2007, Calc. Var.

Exponential convergence to equilibrium

The free energy minimizing equilibrium state of (nFP) is given by

$$f^{\mathrm{eq}}(x) \coloneqq \exp\left(-\frac{\phi(x) - C^{\mathrm{eq}}}{D(x)}\right),$$

where C^{eq} is determined by the normalization $\int_\Omega f dx = \int_\Omega f_0 dx = \int_\Omega f^{\mathrm{eq}} dx = 1$.

Proposition

Let f be a smooth global in time solution of (nFP) *with the initial data f_0 such that*

$$D(x)\log f_0(x) + \phi(x) \in L^\infty(\Omega).$$

Then, we have

$$F[f] - F[f^{\mathrm{eq}}] \le F[f_0]e^{-Ct}, \quad \text{which implies} \quad F[f] - F[f^{\mathrm{eq}}] \to 0 \quad \textit{as} \quad t \to \infty.$$

Here, C depends on d, Ω, f_0, π, D, and ϕ.

Thank you!

References

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See the manuscript for complete literature.